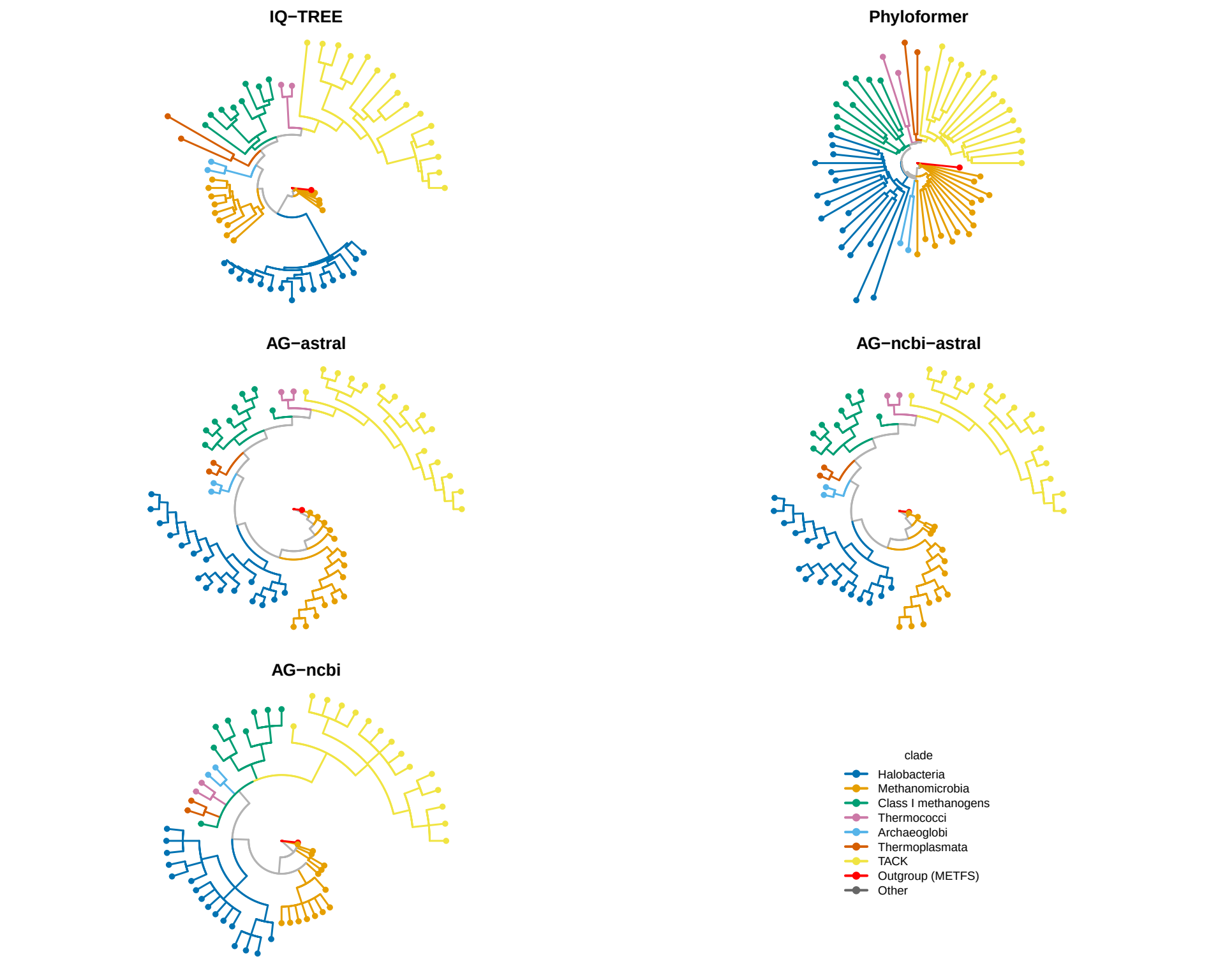
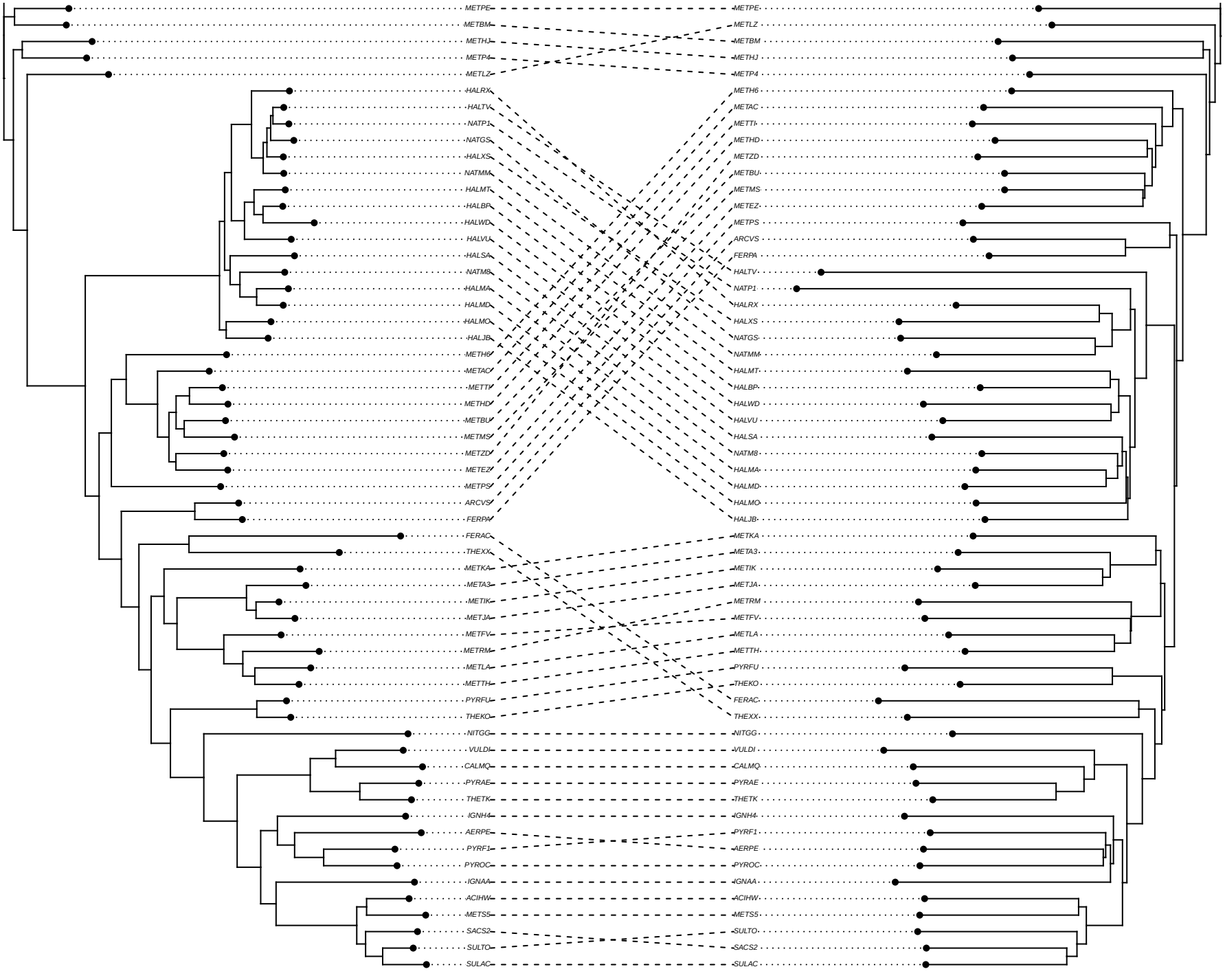
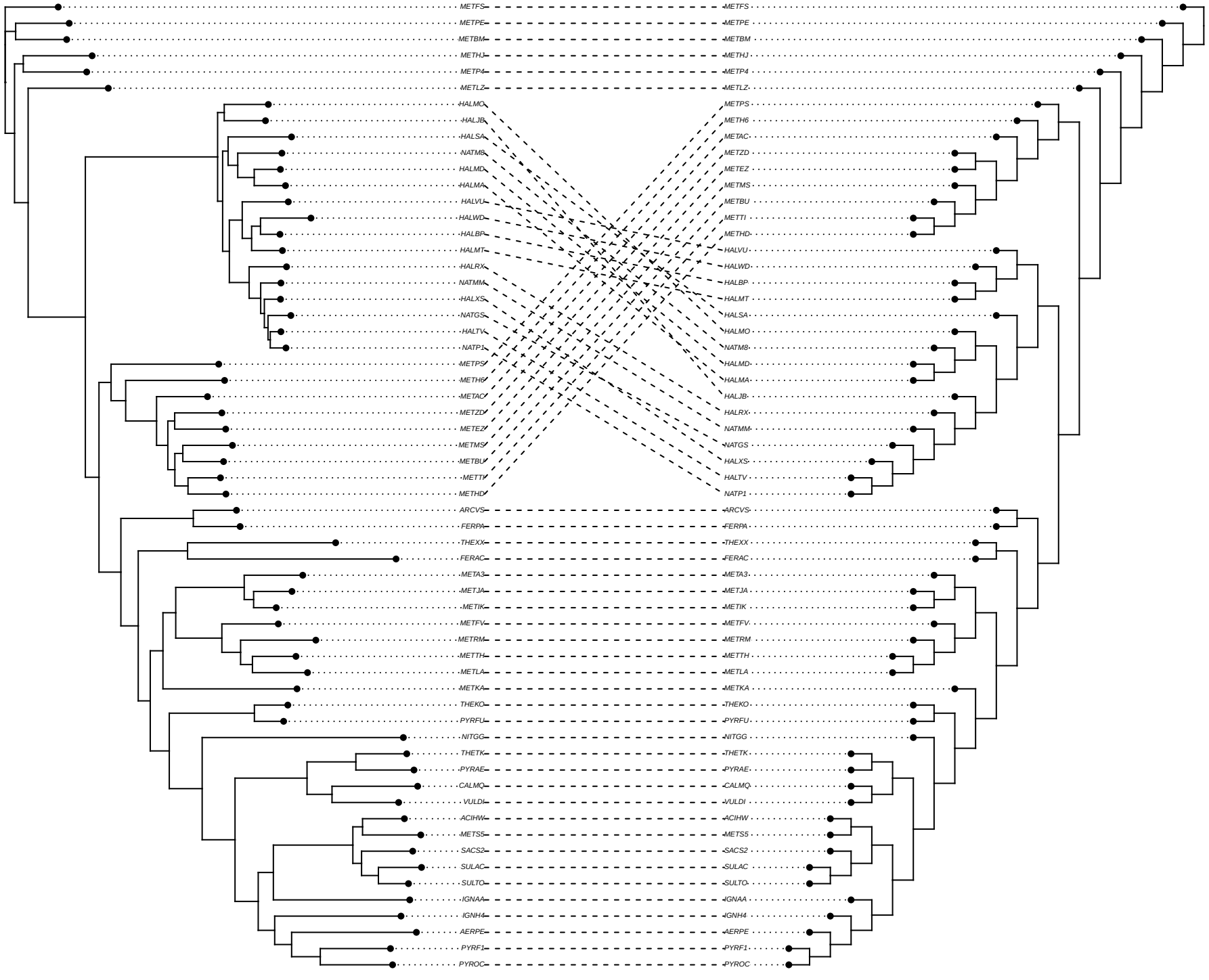

IQ−TREE
Phyloformer
AG−astral
AG−ncbi−astral
AG−ncbi
clade
Halobacteria
Methanomicrobia
Class I methanogens
Thermococci
Archaeoglobi
Thermoplasmata
TACK
Outgroup (METFS)
Other

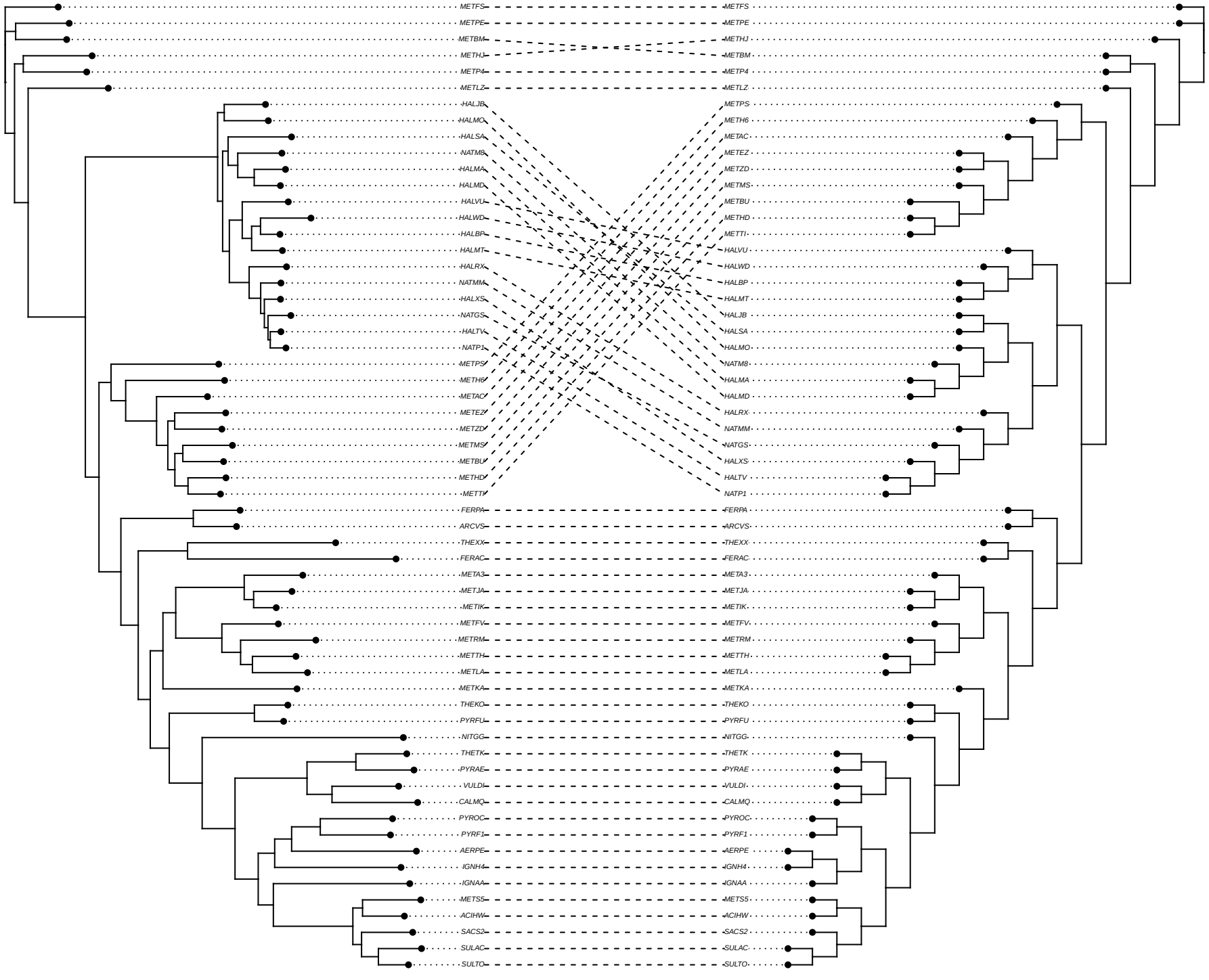
IQ-TREE vs Phylolformer



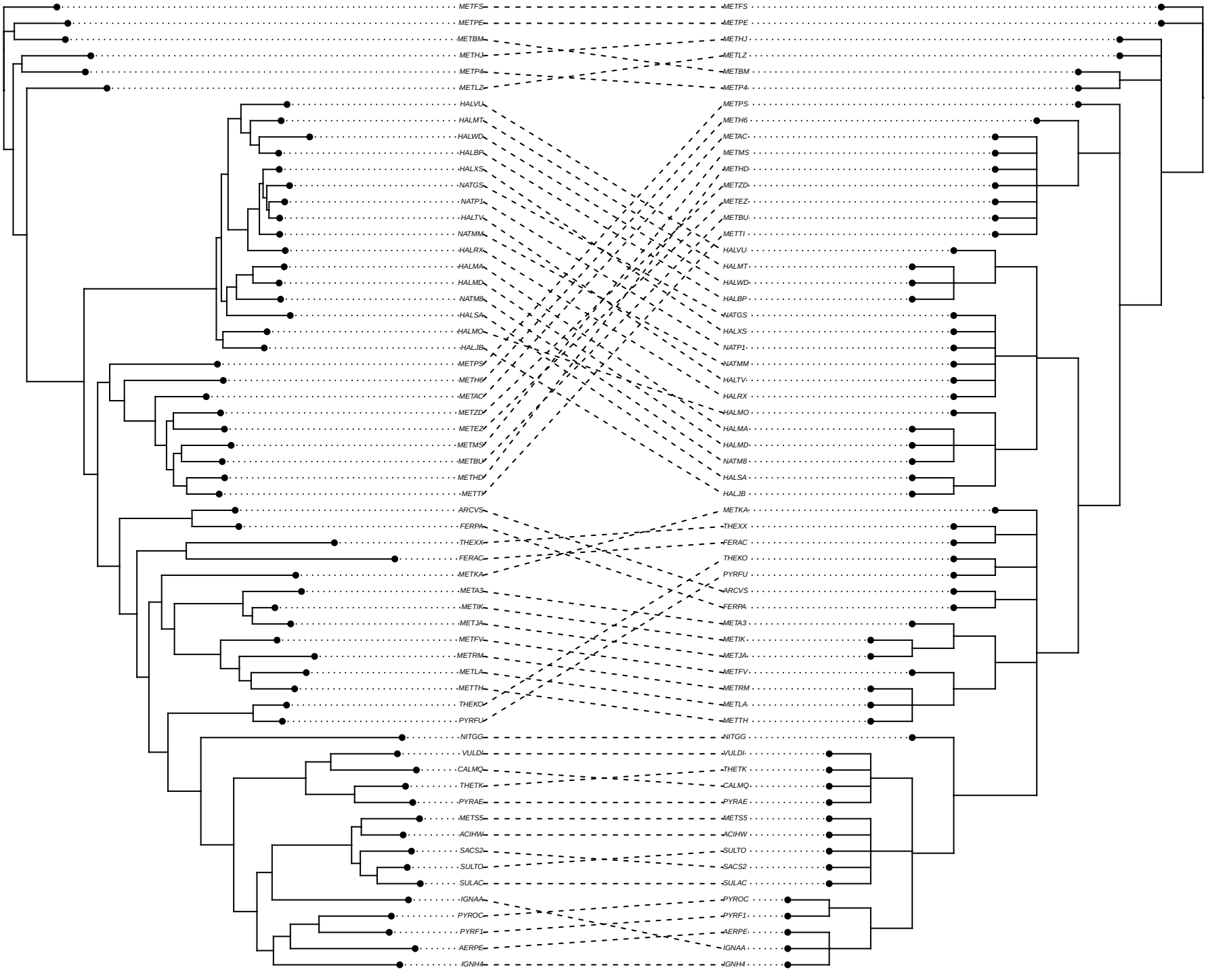
IQ-TREE vs AG-astral

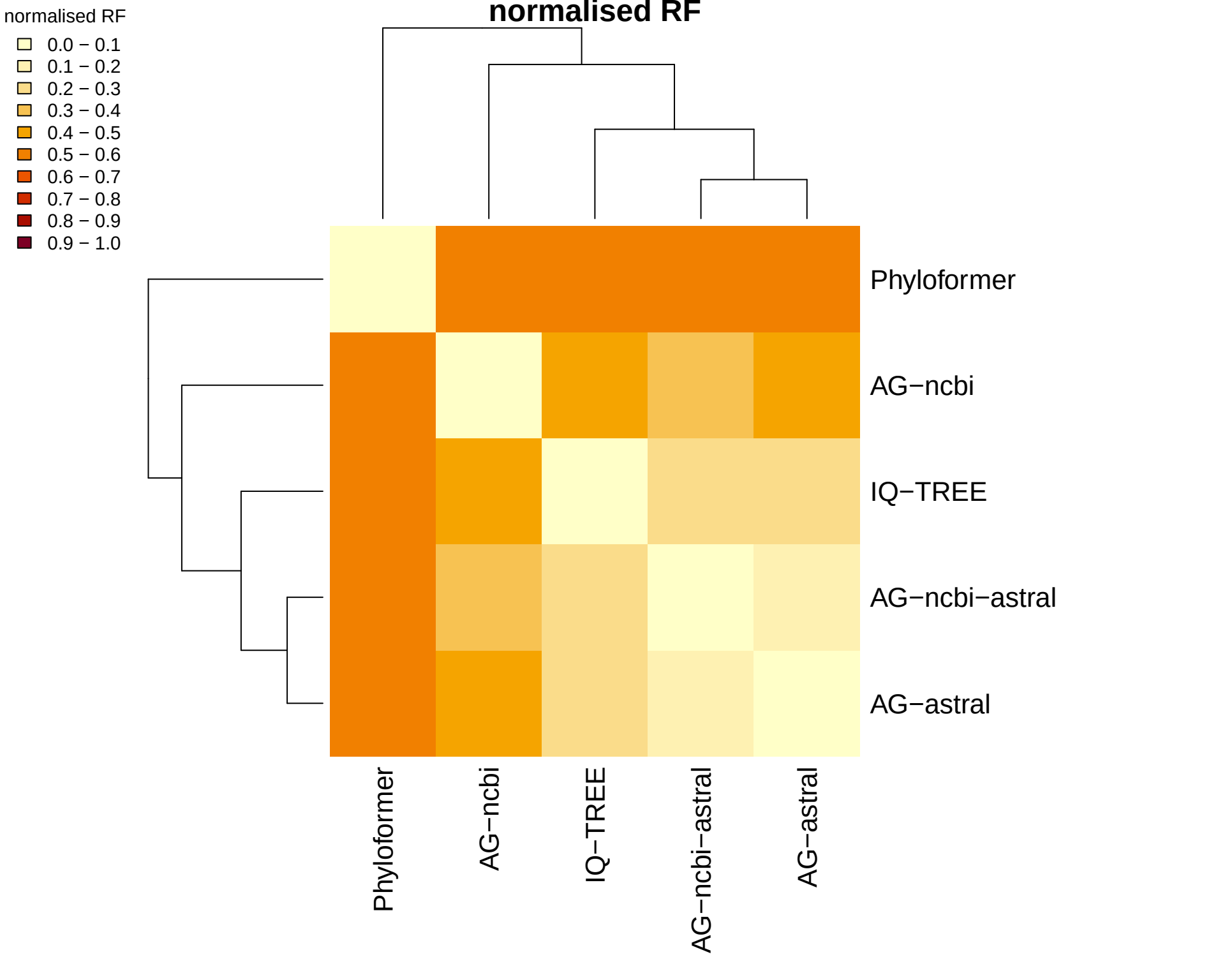


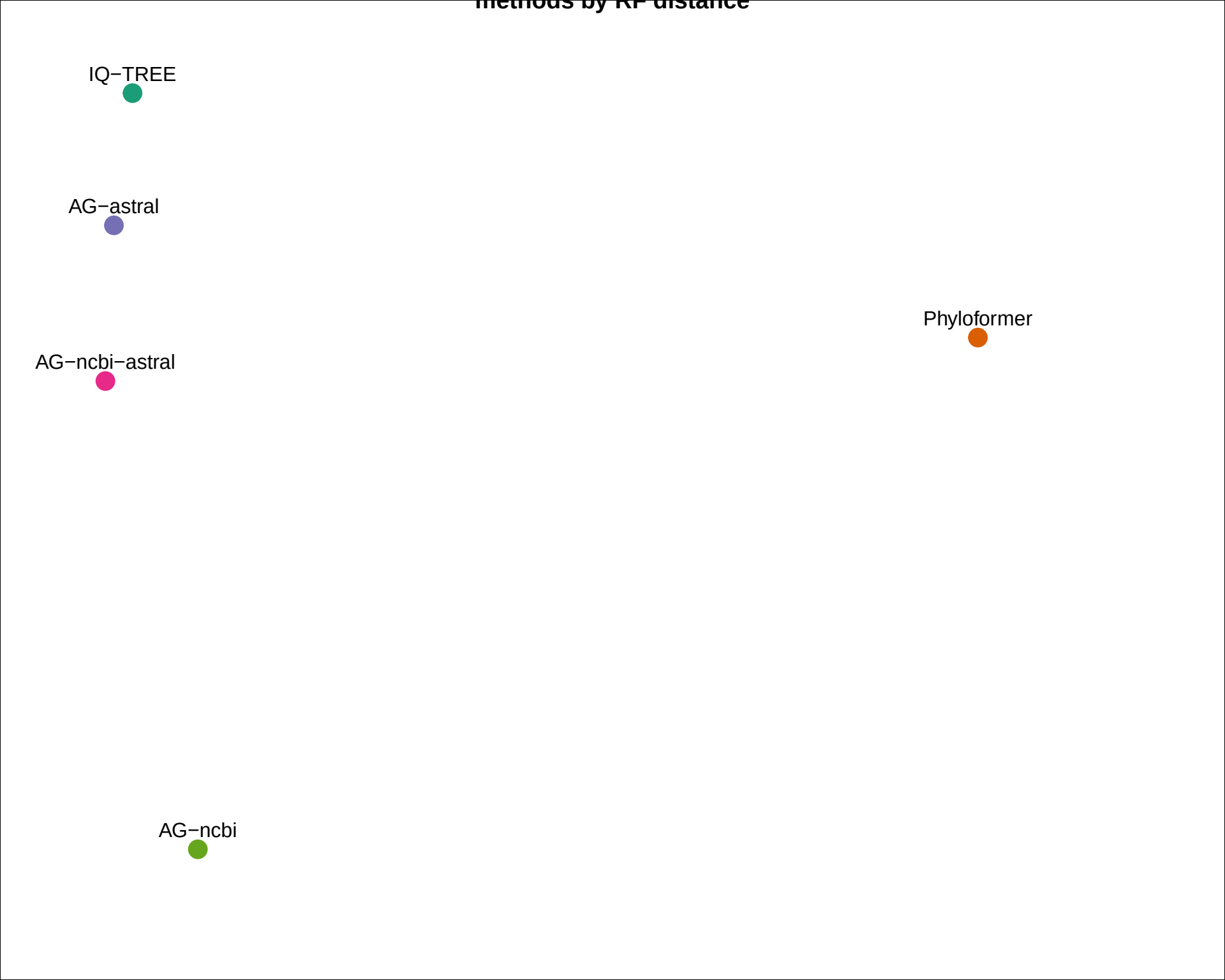
IQ-TREE vs AG-ncbi-astral

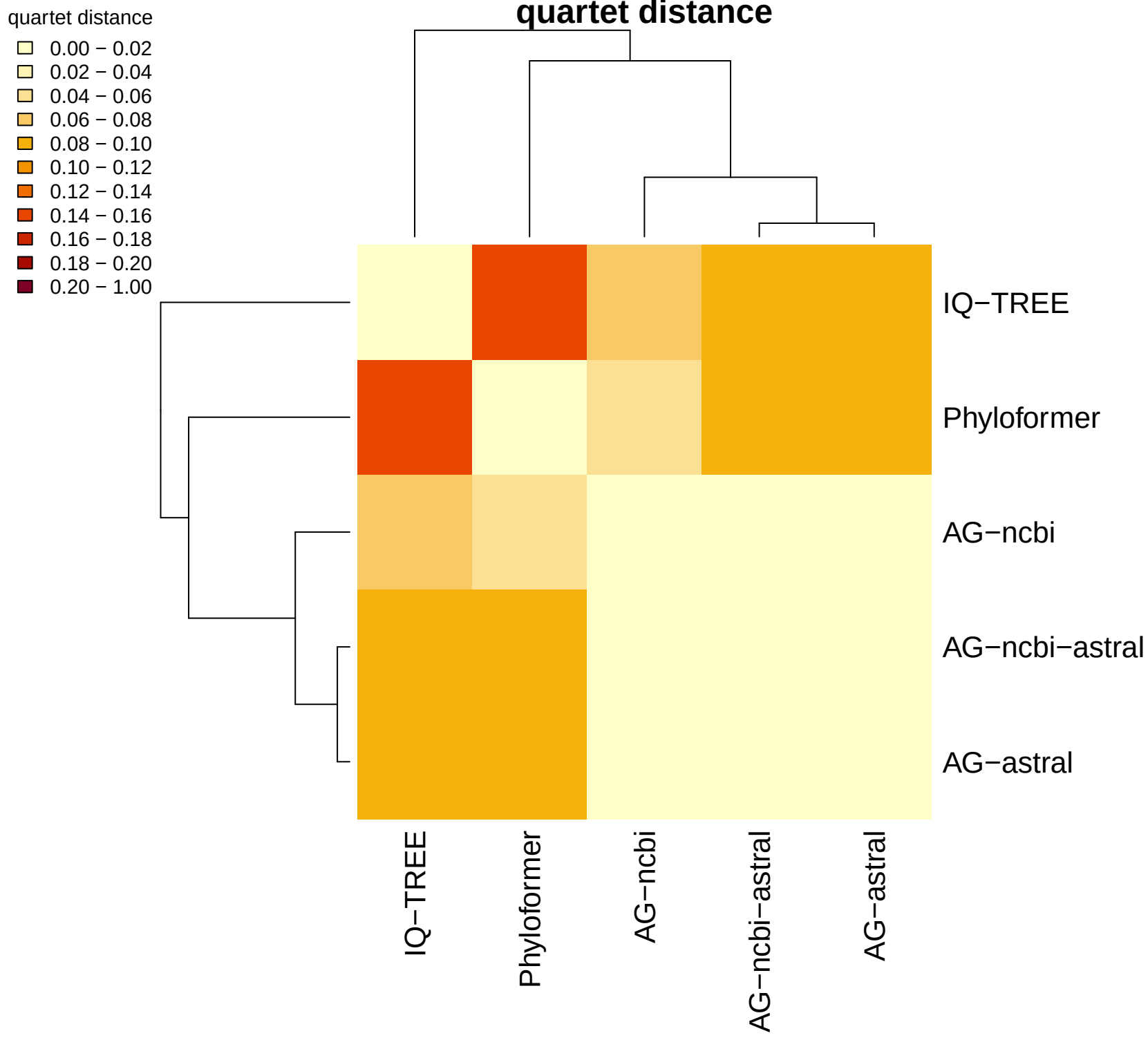


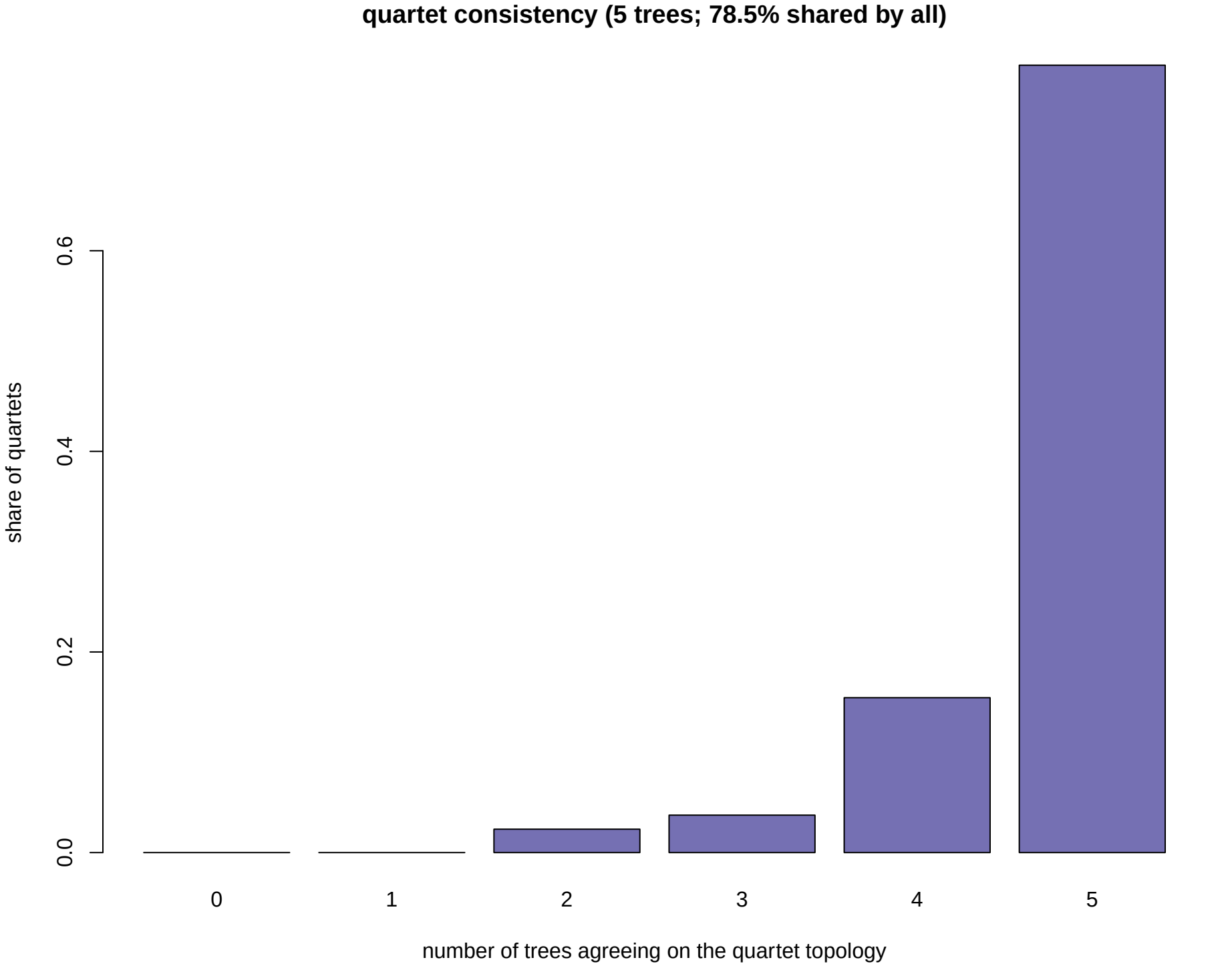
IQ-TREE vs AG-ncbi

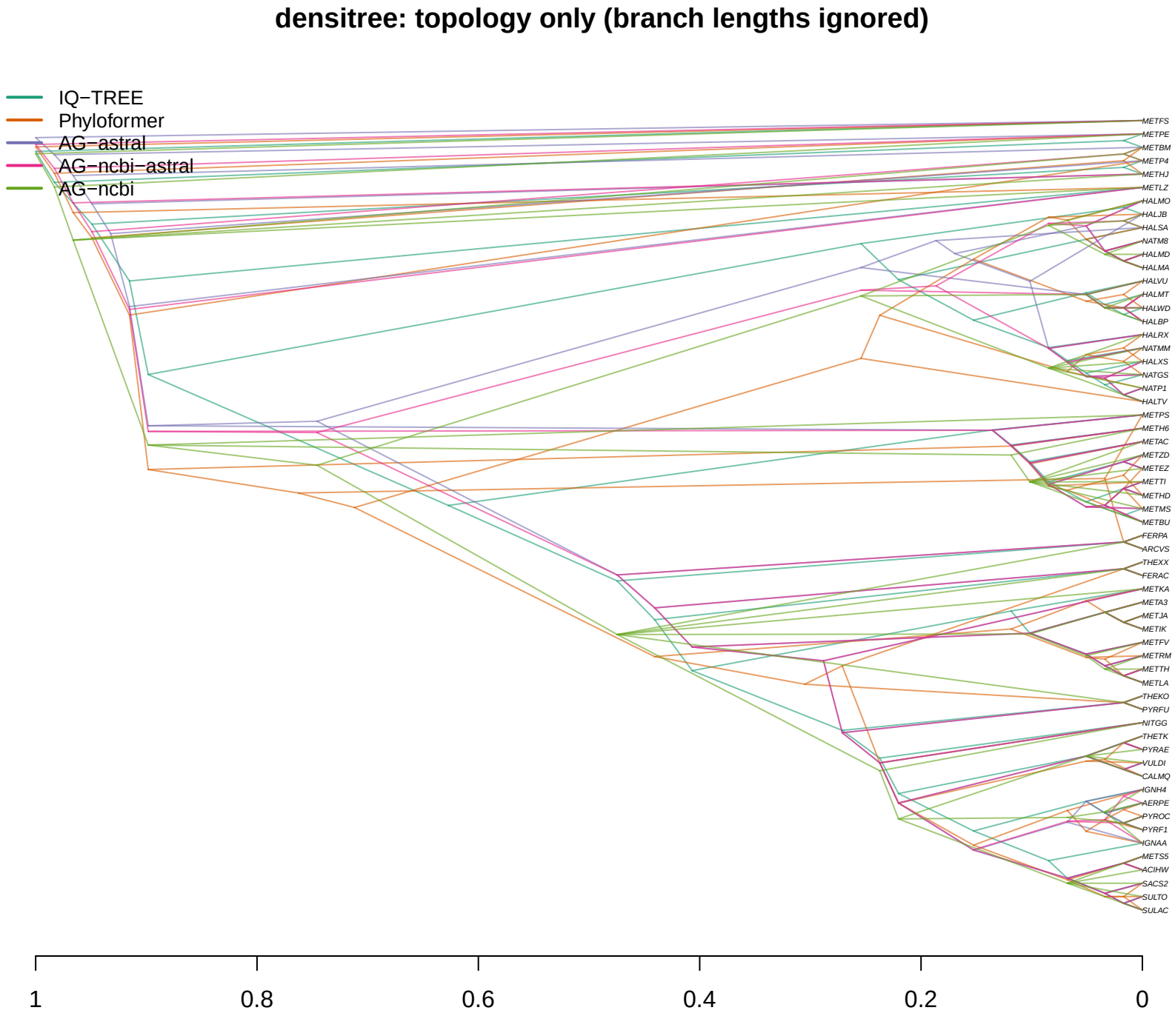


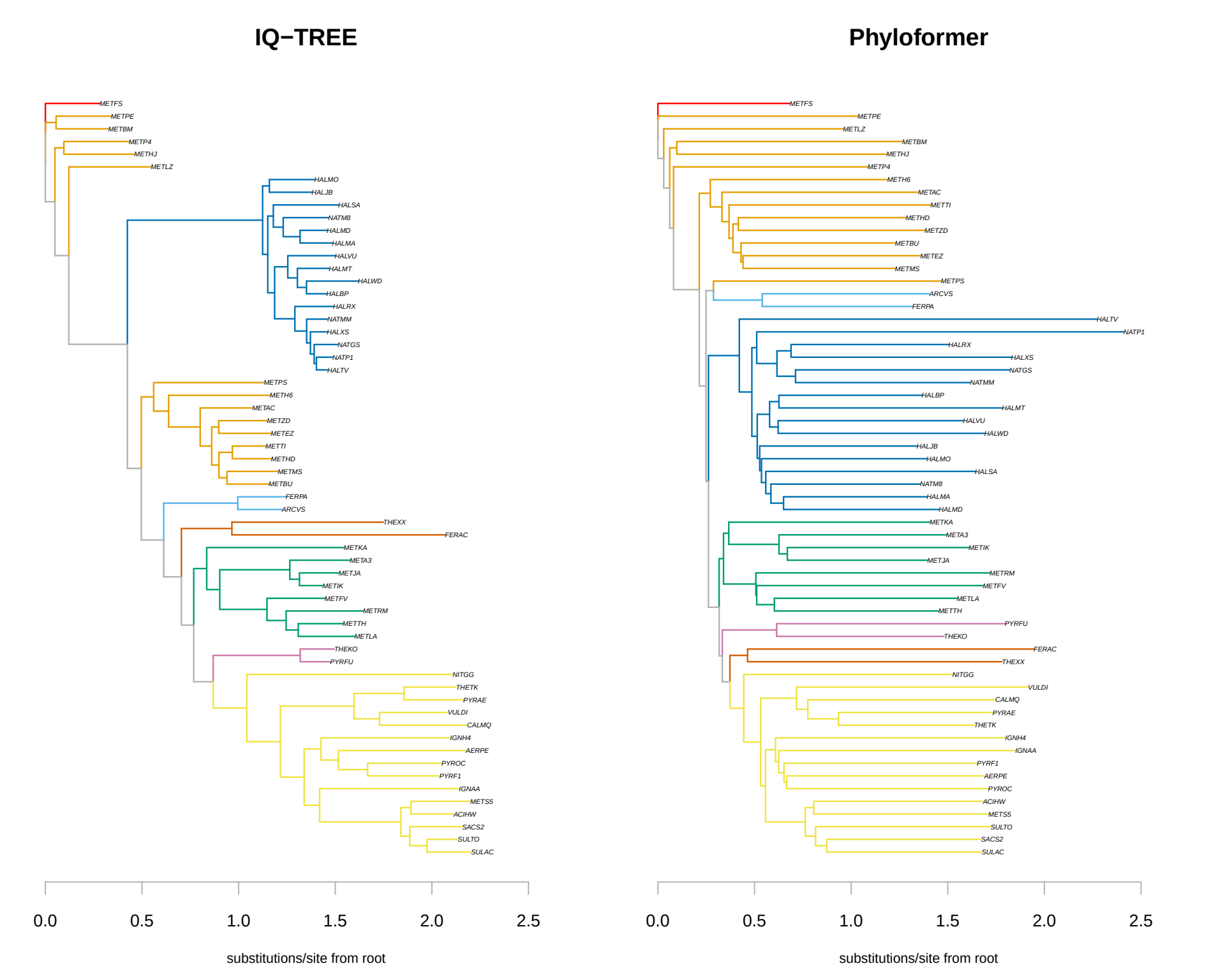

normalised RF
normalised RF
0.0 − 0.1
0.1 − 0.2
0.2 − 0.3
0.3 − 0.4
0.4 − 0.5
0.5 − 0.6
0.6 − 0.7
0.7 − 0.8
0.8 − 0.9
0.9 − 1.0
Phyloformer
AG−ncbi
IQ−TREE
AG−ncbi−astral
AG−astral
Phyloformer
AG−ncbi
IQ−TREE
AG−ncbi−astral
AG−astral


methods by RF distance
IQ−TREE
AG−astral
Phyloformer
AG−ncbi−astral
AG−ncbi

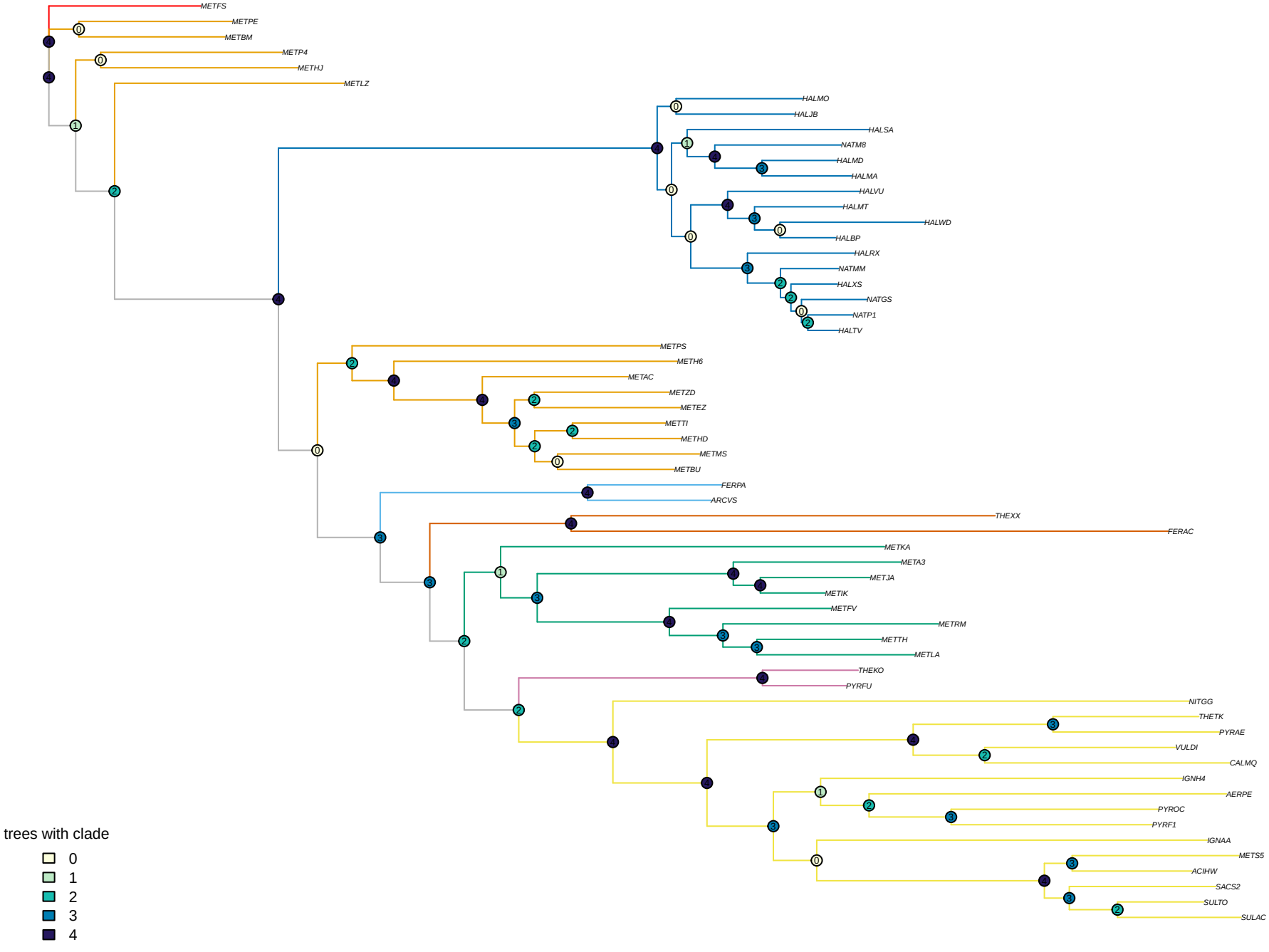

quartet distance
quartet distance
0.00 − 0.02
0.02 − 0.04
0.04 − 0.06
0.06 − 0.08
0.08 − 0.10
0.10 − 0.12
0.12 − 0.14
0.14 − 0.16
0.16 − 0.18
0.18 − 0.20
0.20 − 1.00
IQ−TREE
Phyloformer
AG−ncbi
AG−ncbi−astral
AG−astral
IQ−TREE
Phyloformer
AG−ncbi
AG−ncbi−astral
AG−astral


quartet consistency (5 trees; 78.5% shared by all)
share of quartets
0.0
0.2
0.4
0.6
0
1
2
3
4
5
number of trees agreeing on the quartet topology

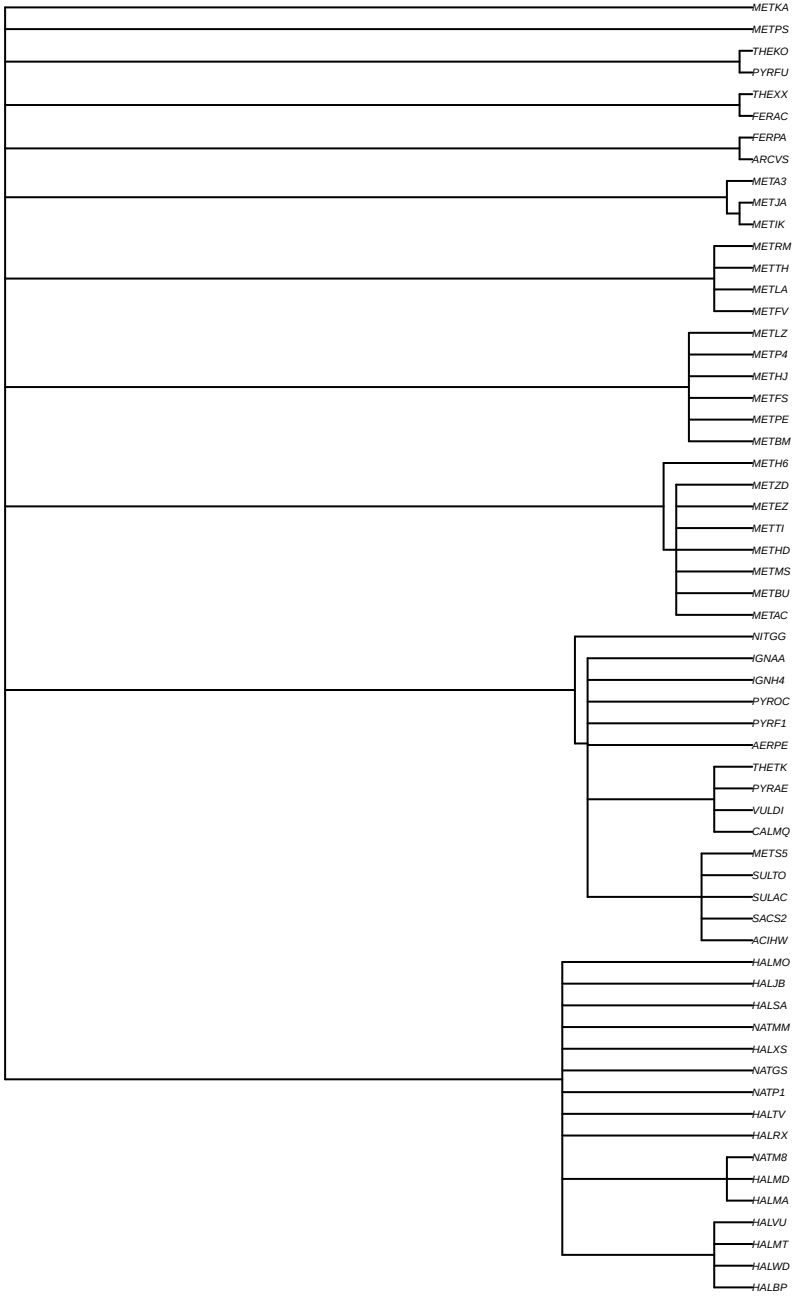

densitree: topology only (branch lengths ignored)
IQ−TREE
Phyloformer
AG−astral
AG−ncbi−astral
AG−ncbi
METFS
METPE
METBM
METP4
METHJ
METLZ
HALMO
HALJB
HALSA
NATM8
HALMD
HALMA
HALVU
HALMT
HALWD
HALBP
HALRX
NATMM
HALXS
NATGS
NATP1
HALTV
METPS
METH6
METAC
METZD
METEZ
METTI
METHD
METMS
METBU
FERPA
ARCVS
THEXX
FERAC
METKA
META3
METJA
METIK
METFV
METRM
METTH
METLA
THEKO
PYRFU
NITGG
THETK
PYRAE
VULDI
CALMQ
IGNH4
AERPE
PYROC
PYRF1
IGNAA
METS5
ACIHW
SACS2
SULTO
SULAC
1
0.8
0.6
0.4
0.2
0


IQ−TREE
METFS
METPE
METBM
METP4
METHJ
METLZ
HALMO
HALJB
HALSA
NATM8
HALMD
HALMA
HALVU
HALMT
HALWD
HALBP
HALRX
NATMM
HALXS
NATGS
NATP1
HALTV
METPS
METH6
METAC
METZD
METEZ
METTI
METHD
METMS
METBU
FERPA
ARCVS
THEXX
FERAC
METKA
META3
METJA
METIK
METFV
METRM
METTH
METLA
THEKO
PYRFU
NITGG
THETK
PYRAE
VULDI
CALMQ
IGNH4
AERPE
PYROC
PYRF1
IGNAA
METS5
ACIHW
SACS2
SULTO
SULAC
0.0
0.5
1.0
1.5
2.0
2.5
substitutions/site from root
Phyloformer
METFS
METPE
METLZ
METBM
METHJ
METP4
METH6
METAC
METTI
METHD
METZD
METBU
METEZ
METMS
METPS
ARCVS
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HALTV
NATP1
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VULDI
CALMQ
PYRAE
THETK
IGNH4
IGNAA
PYRF1
AERPE
PYROC
ACIHW
METS5
SULTO
SACS2
SULAC
0.0
0.5
1.0
1.5
2.0
2.5
substitutions/site from root

IQ−TREE clades: number of the other 4 trees that contain each clade



strict



majority rule

